



Premier University Chittagong

Department of Computer Science & Engineering

Assignment

Assignment No	03
Assignment Name	Alternative of CT-3
Course Code	CSE 2221
Course Title	Database Management System
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Answer to the Question no 1:

Corrected 1NF Table (atomic Exam values)

Every exam becomes its own row because the key includes Exam.

Patient Num	AppointmentDate	Exam	PatientName	Patient DOB	Dentist Num	Dentist Name	ClinicName	ClinicAddress	CompleteFlag
0127	03/09/2023	Dental	Sandra Ho	13/05/78	001	Carl	Smile Bright	George St	Y
0127	03/09/2023	Periodontal	Sandra Ho	13/05/78	001	Carl	Smile Bright	George St	Y
0127	10/10/2023	Panoramic	Sandra Ho	13/05/78	001	Carl	Tooth Fairy	Thomas St	Y
0127	20/12/2023	Tooth decay	Sandra Ho	13/05/78	001	Carl	Pearly White	Beacon Ave	Null
0127	20/12/2023	Bitewing	Sandra Ho	13/05/78	001	Carl	Pearly White	Beacon Ave	Null
0823	03/09/2023	Orthodontal	Jeff Daniels	07/08/67	002	Sonia	Smile Bright	George St	Null
0823	03/09/2023	Periapical Cyst	Jeff Daniels	07/08/67	002	Sonia	Smile Bright	George St	Null
0823	03/09/2023	Bitewing	Jeff Daniels	07/08/67	002	Sonia	Tooth Fairy	Thomas St	Y
0146	04/10/2023	Periodontal	David Falkner	16/06/79	003	Brad	Tooth Fairy	Thomas St	Y
0146	04/10/2023	Recall	David Falkner	16/06/79	003	Brad	Tooth Fairy	Thomas St	Y
0239	05/06/2023	Cleaning	Will Bosworth	23/03/98	004	Carl	Tooth Fairy	Thomas St	Null
0749	09/11/2023	X-Rays	Lynn Fields	17/07/03	005	Brad	Pearly White	Beacon Ave	Y

Changes made

- ❖ Split multivalued Exam into individual rows.
- ❖ Replaced missing values with null.
- ❖ Removed implied repeating values and explicitly entered all known data.
- ❖ Ensured table is rectangular and all attributes atomic.

Answer to the Question no 2:

Unnormalized Relation:

DentalExam (PatientNum, AppointmentDate, Exam, PatientName, PatientDOB, DentistNum, DentistName, ClinicName, ClinicAddress, CompleteFlag)

Primary Key: (PatientNum, AppointmentDate, Exam)

Answer to the Question no 3:

Functional Dependencies

1. PatientNum → PatientName, PatientDOB
2. DentistNum → DentistName
3. ClinicName → ClinicAddress
4. PatientNum → DentistNum, DentistName
5. (PatientNum, AppointmentDate, Exam) → PatientName, PatientDOB, DentistNum, DentistName, ClinicName, ClinicAddress, CompleteFlag

Answer to the Question no 4:

1NF → 2NF Relations (Removed partial dependencies)

Patient (PatientNum, PatientName, PatientDOB, DentistNum*)

Appointment (PatientNum, AppointmentDate, ClinicName, ClinicAddress, DentistNum*)

DentalExam (PatientNum, AppointmentDate, Exam, CompleteFlag)

Answer to the Question no 5:

2NF → 3NF Relations (Removed transitive dependencies)

Patient (PatientNum, PatientName, PatientDOB, DentistNum*)

Dentist (DentistNum, DentistName)

Clinic (ClinicName, ClinicAddress)

Appointment (PatientNum, AppointmentDate, ClinicName*)

DentalExam (PatientNum, AppointmentDate, Exam, CompleteFlag)

Answer to the Question no 6:

3NF → BCNF

Already in BCNF (all determinants are keys in their relations).

No further decomposition needed.

Patient (PatientNum, PatientName, PatientDOB, DentistNum*)

Dentist (DentistNum, DentistName)

Clinic (ClinicName, ClinicAddress)

Appointment (PatientNum, AppointmentDate, ClinicName*)

DentalExam (PatientNum, AppointmentDate, Exam, CompleteFlag)

Answer to the Question no 7:

The new relation:

DentalExam (PatientNum, AppointmentDate, Exam, ClinicNum*, CompleteFlag,

AppointmentNum)

AppointmentNum is not part of the key → causes anomalies. A single appointment should have its own primary key.

Create separate APPOINTMENT table:

Appointment (AppointmentNum, PatientNum*, AppointmentDate, DentistNum*, ClinicName*)

DentalExam (AppointmentNum, Exam, CompleteFlag)

Answer to the Question no 8:

Final Normalized Relations (Adding surrogate keys where needed)

Patient (PatientNum, PatientName, PatientDOB, DentistNum*)

Dentist (DentistNum, DentistName)

Clinic (ClinicNum, ClinicName, ClinicAddress) **[ClinicNum is a surrogate key for ClinicName.]**

Appointment (AppointmentNum, PatientNum*, AppointmentDate, DentistNum*, ClinicNum*)

[AppointmentNum is the surrogate key for the appointment.]

DentalExam (AppointmentNum, Exam, CompleteFlag)

The Normalized Relations in BCNF
